

Linear System Fall 2026

Homework One

Due: Friday, September 18

Exercise 1 Consider a system described by a single n^{th} -order linear differential equation of the form

$$q^{(n)} + a_{n-1}q^{(n-1)} + \cdots + a_1\dot{q} + a_0q = u$$

where $q(t) \in \mathbb{R}$ and $q^{(n)} := \frac{d^n q}{dt^n}$. By appropriate definition of state variables, obtain a first order state space description of this system.

Exercise 2 By appropriate definition of state variables, obtain a first order state space description of the following systems where q_1 and q_2 are real scalars.

(i)

$$\begin{aligned} 2\ddot{q}_1 + \ddot{q}_2 + \sin q_1 &= 0, \\ \dot{q}_1 + 2\dot{q}_2 + \sin q_2 &= 0. \end{aligned}$$

(ii)

$$\begin{aligned} \ddot{q}_1 + \dot{q}_2 + q_1^3 &= 0, \\ \dot{q}_1 + \dot{q}_2 + q_2^2 &= 0. \end{aligned}$$

Exercise 3 Obtain a state-space description of the following system.

$$\begin{aligned} \ddot{q}_1 + \ddot{q}_2 + \sin q_1 &= u, \\ \dot{q}_2 + q_1 + q_2 &= 0, \\ y &= q_1 + q_2. \end{aligned}$$

Exercise 4 Consider the second order difference equation

$$q(k+2) = q(k+1) + q(k)$$

Obtain a state space description of this system.

Exercise 5 Obtain a state space representation of the following system:

$$q(k+n) + a_{n-1}q(k+n-1) + \cdots + a_1q(k+1) + a_0q(k) = 0$$

where $q(k) \in \mathbb{R}$.

Exercise 6 Obtain a state description of the following system:

$$\begin{aligned}q_1(k+2) + q_2(k+1) + q_1(k) &= u(k) \\q_1(k+2) - q_2(k+1) + q_2(k) &= 0 \\y(k) &= q_1(k+1) + q_2(k)\end{aligned}$$

Exercise 7 Recall the two pendulum cart example we mentioned in the class. Consider the following parameter sets:

	m_0	m_1	m_2	l_1	l_2	g
P1	2	1	1	1	1	1
P2	2	1	1	1	0.99	1
P3	2	1	0.5	1	1	1
P4	2	1	1	1	0.5	1

and initial conditions,

	y	θ_1	θ_2	$\dot{y}$	$\dot{\theta}_1$	$\dot{\theta}_2$
IC1	0	-10°	10°	0	0	0
IC2	0	10°	10°	0	0	0
IC3	0	-90°	90°	0	0	0
IC4	0	-90.01°	90°	0	0	0
IC5	0	100°	100°	0	0	0
IC6	0	100.01°	100°	0	0	0
IC7	0	179.99°	0°	0	0	0

Simulate (use matlab/simulink or python or anything you want to code it) the system with $u = 0$ using the following combinations:

$$P1 : \quad IC1, IC2, IC3, IC7$$

$$P4 : \quad IC1, IC2, IC3, IC4$$